

CORE CSE SERIES — MCQ EDITION

Operating Systems MCQ Handbook

Basic to Advanced Multiple-Choice Questions for Company OTs, PYQ-style Practice & Core Placement Preparation — with Answers & Explanations

Process Mgmt

CPU Scheduling

Sync & Deadlocks

Memory Mgmt

File Systems

Threads

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Topic-wise MCQs with Explanations

1. Basics, Process Management & PCB Process states, PCB, context switching, system calls

Q1 What is the primary function of an Operating System?

BASIC

- A. To manage hardware resources and provide an interface between user and hardware**
- B. To compile source code
- C. To design websites
- D. To manage only network connections

Correct Answer: A. To manage hardware resources and provide an interface between user and hardware

Explanation: The OS manages hardware resources (CPU, memory, I/O) and acts as an intermediary between applications and hardware.

Q2 Which of the following is NOT a state of a process in the standard 5-state process model?

BASIC

- A. New
- B. Ready
- C. Running
- D. Compiled**

Correct Answer: D. Compiled

Explanation: The standard states are New, Ready, Running, Waiting/Blocked, and Terminated. 'Compiled' is not a process state.

Q3 What does PCB stand for in operating systems?

BASIC

- A. Process Control Block**
- B. Program Control Buffer
- C. Process Cache Block
- D. Program Compilation Block

Correct Answer: A. Process Control Block

Explanation: The Process Control Block stores all information about a process — PID, state, registers, memory limits, scheduling info — needed for context switching.

Q4 Context switching refers to:

BASIC

- A. Switching the CPU from one process to another by saving/restoring state**
- B. Switching between two hard disks
- C. Converting high-level code to machine code
- D. Switching between two users on the same terminal

Correct Answer: A. Switching the CPU from one process to another by saving/restoring state

Explanation: Context switching saves the state (registers, PC) of the current process into its PCB and loads the state of the next process to be run.

Q5 Which system call is used to create a new process in UNIX/Linux?

INTERMEDIATE

- A. fork()**
- B. create()
- C. new_process()
- D. spawn()

Correct Answer: A. fork()

Explanation: fork() creates a new process (child) that is a near-copy of the calling (parent) process.

Q6 A process that has finished execution but still has an entry in the process table is called a:

INTERMEDIATE

- A. Zombie process**
- B. Orphan process
- C. Daemon process
- D. Ready process

Correct Answer: A. Zombie process

Explanation: A zombie process has completed execution but its exit status hasn't been read by the parent via wait(), so its PCB entry remains.

Q7 An orphan process is one whose:

INTERMEDIATE

- A. Parent has terminated before it, and it gets adopted by init/systemd**
- B. Child has terminated
- C. Memory has been deallocated
- D. PCB has been deleted

Correct Answer: A. Parent has terminated before it, and it gets adopted by init/systemd

Explanation: When a parent terminates before its child, the child becomes an orphan and is typically re-parented to the init process (PID 1).

Q8 Which of these is a long-term scheduler's responsibility?

ADVANCED

- A. Deciding which processes are admitted into the ready queue (job scheduling)
- B. Deciding which ready process runs next on CPU
- C. Swapping processes in and out of memory
- D. Handling interrupts

Correct Answer: A. Deciding which processes are admitted into the ready queue (job scheduling)

Explanation: The long-term (job) scheduler controls the degree of multiprogramming by deciding which jobs are brought into the ready queue from the job pool.

Q9 The medium-term scheduler is primarily responsible for:

ADVANCED

- A. Swapping processes in/out of main memory to control multiprogramming
- B. Assigning CPU to processes
- C. Creating new processes
- D. Managing file access

Correct Answer: A. Swapping processes in/out of main memory to control multiprogramming

Explanation: The medium-term scheduler suspends (swaps out) and resumes processes to manage the degree of multiprogramming, especially under memory pressure.

2. CPU Scheduling FCFS, SJF, Round Robin, Priority, multilevel queues

Q10 Which scheduling algorithm can lead to the 'convoy effect'?

BASIC

- A. First Come First Served (FCFS)
- B. Round Robin
- C. Shortest Job First
- D. Priority Scheduling with aging

Correct Answer: A. First Come First Served (FCFS)

Explanation: In FCFS, a long process at the front of the queue makes all shorter processes behind it wait — this is the convoy effect.

Q11 Which scheduling algorithm is provably optimal for minimizing average waiting time (non-preemptive)? INTERMEDIATE

- A. Shortest Job First (SJF)**
- B. FCFS
- C. Round Robin
- D. Priority Scheduling

Correct Answer: A. Shortest Job First (SJF)

Explanation: SJF (non-preemptive) is proven to give the minimum average waiting time among non-preemptive algorithms, assuming burst times are known.

Q12 In Round Robin scheduling, performance is highly sensitive to: BASIC

- A. The size of the time quantum**
- B. The number of processes only
- C. The arrival order of processes
- D. The priority values

Correct Answer: A. The size of the time quantum

Explanation: A very small time quantum increases context-switch overhead; a very large one makes RR behave like FCFS.

Q13 What is 'starvation' in the context of CPU scheduling? BASIC

- A. A low-priority process waits indefinitely because higher-priority processes keep arriving**
- B. A process uses too much memory
- C. A process crashes due to lack of input
- D. The CPU runs out of instructions

Correct Answer: A. A low-priority process waits indefinitely because higher-priority processes keep arriving

Explanation: Starvation occurs when a process is perpetually denied CPU time because the scheduler keeps favoring other (typically higher priority) processes.

Q14 Which technique is commonly used to prevent starvation in priority scheduling? INTERMEDIATE

- A. Aging (gradually increasing priority of waiting processes)**
- B. Reducing time quantum
- C. Increasing process priority randomly
- D. Disabling interrupts

Correct Answer: A. Aging (gradually increasing priority of waiting processes)

Explanation: Aging gradually increases the priority of processes that have been waiting a long time, ensuring they eventually get scheduled.

Q15 Preemptive SJF is also known as:

INTERMEDIATE

- A. Shortest Remaining Time First (SRTF)**
- B. Longest Job First
- C. Multilevel Feedback Queue
- D. Round Robin

Correct Answer: A. Shortest Remaining Time First (SRTF)

Explanation: SRTF preempts the currently running process if a new process arrives with a shorter remaining burst time than the current process.

Q16 In Multilevel Feedback Queue scheduling, a process can:

ADVANCED

- A. Move between different priority queues based on its behavior/CPU usage**
- B. Only stay in one queue for its entire life
- C. Never be preempted
- D. Only run once

Correct Answer: A. Move between different priority queues based on its behavior/CPU usage

Explanation: MLFQ allows processes to move between queues — e.g., a CPU-bound process may be demoted to a lower-priority queue, while I/O-bound processes stay higher priority.

Q17 Turnaround time of a process is defined as:

BASIC

- A. Completion time – Arrival time**
- B. Completion time – Burst time
- C. Waiting time + Arrival time
- D. Burst time – Waiting time

Correct Answer: A. Completion time – Arrival time

Explanation: Turnaround time = Completion Time – Arrival Time = Waiting Time + Burst Time.

Q18 If Round Robin uses an infinitely large time quantum, it behaves like:

INTERMEDIATE

- A. FCFS**
- B. SJF
- C. Priority Scheduling
- D. Multilevel Queue

Correct Answer: A. FCFS

Explanation: With a very large quantum, each process runs to completion before the next starts, which is identical to FCFS behavior.

3. Process Synchronization & Deadlocks Critical section, semaphores, mutex, deadlock

conditions

Q19 The Critical Section Problem requires a solution that satisfies which three properties?

INTERMEDIATE

- A. Mutual Exclusion, Progress, Bounded Waiting
- B. Atomicity, Isolation, Durability
- C. Consistency, Availability, Partition Tolerance
- D. Deadlock, Starvation, Livelock

Correct Answer: A. Mutual Exclusion, Progress, Bounded Waiting

Explanation: A valid critical-section solution must ensure mutual exclusion, progress (no indefinite postponement of entry decisions), and bounded waiting (a limit on how long a process waits).

Q20 A binary semaphore can take values:

BASIC

- A. 0 and 1 only
- B. Any integer
- C. Only positive integers
- D. Only negative integers

Correct Answer: A. 0 and 1 only

Explanation: A binary semaphore (similar to a mutex) is restricted to values 0 (locked) and 1 (unlocked).

Q21 Which of these is NOT one of the four necessary conditions for deadlock?

INTERMEDIATE

- A. Preemption
- B. Mutual Exclusion
- C. Hold and Wait
- D. Circular Wait

Correct Answer: A. Preemption

Explanation: The four Coffman conditions are Mutual Exclusion, Hold and Wait, No Preemption (not Preemption), and Circular Wait.

Q22 The Banker's Algorithm is used for:

ADVANCED

- A. Deadlock avoidance**
- B. Deadlock detection
- C. CPU scheduling
- D. Page replacement

Correct Answer: A. Deadlock avoidance

Explanation: The Banker's Algorithm checks if granting a resource request keeps the system in a 'safe state' to avoid entering deadlock.

Q23 A 'safe state' in the Banker's algorithm means:

ADVANCED

- A. There exists at least one sequence in which all processes can complete without deadlock**
- B. No process is currently waiting
- C. All resources are free
- D. The system has zero processes

Correct Answer: A. There exists at least one sequence in which all processes can complete without deadlock

Explanation: A safe state guarantees a safe sequence exists, so even if all processes request their maximum claims, deadlock can still be avoided.

Q24 Which of the following can break the 'Hold and Wait' condition to prevent deadlock?

ADVANCED

- A. Requiring processes to request all needed resources at once, before execution**
- B. Allowing processes to hold multiple resource types simultaneously
- C. Increasing the number of resource instances
- D. Using priority scheduling

Correct Answer: A. Requiring processes to request all needed resources at once, before execution

Explanation: Forcing processes to request all resources upfront (or release held resources before requesting more) breaks the hold-and-wait condition.

Q25 What is a race condition?

BASIC

- A. When the outcome of concurrent execution depends on the timing/order of execution**
- B. When two processes deadlock
- C. When a process runs out of memory
- D. When CPU utilization is 100%

Correct Answer: A. When the outcome of concurrent execution depends on the timing/order of execution

Explanation: A race condition occurs when multiple processes/threads access shared data concurrently and the final result depends on the relative timing of their execution.

Q26 The 'Dining Philosophers' problem is a classic example used to illustrate:

INTERMEDIATE

- A. Deadlock and synchronization challenges**
- B. Memory fragmentation
- C. Page replacement policies
- D. Disk scheduling

Correct Answer: A. Deadlock and synchronization challenges

Explanation: It models resource contention (forks) among concurrent processes (philosophers) and illustrates deadlock, starvation, and synchronization.

Q27 What is a spinlock?

ADVANCED

- A. A lock where a thread repeatedly checks (busy-waits) until the lock becomes available**
- B. A lock that automatically unlocks after a timer
- C. A lock used only for disk I/O
- D. A lock that only one CPU core can use

Correct Answer: A. A lock where a thread repeatedly checks (busy-waits) until the lock becomes available

Explanation: A spinlock causes a thread to loop (spin) checking a condition rather than sleeping, which is efficient only for very short critical sections.

Q28 Which of these is TRUE about mutex vs semaphore?

INTERMEDIATE

- A. A mutex provides ownership (only the locking thread can unlock); a semaphore does not**
- B. A semaphore can only take values 0 or 1
- C. A mutex allows multiple threads simultaneously
- D. There is no difference between them

Correct Answer: A. A mutex provides ownership (only the locking thread can unlock); a semaphore does not

Explanation: Mutex enforces ownership — only the thread that locked it can unlock it. Counting semaphores allow a set number of concurrent accesses and have no ownership concept.

4. Memory Management & Virtual Memory Paging, segmentation, page replacement,

thrashing

Q29 What is internal fragmentation?

BASIC

- A. Wasted space within an allocated fixed-size block/page that isn't fully used**
- B. Wasted space between allocated blocks in memory
- C. Fragmentation of a hard disk
- D. Fragmentation of network packets

Correct Answer: A. Wasted space within an allocated fixed-size block/page that isn't fully used

Explanation: Internal fragmentation occurs when a fixed-size partition/page is allocated but the process doesn't use all of it, wasting the remaining space within the block.

Q30 What is external fragmentation?

BASIC

- A. Free memory is split into small, non-contiguous blocks, none large enough for a new request**
- B. A process crashes due to a segmentation fault
- C. A CPU cache miss
- D. A disk block being reused

Correct Answer: A. Free memory is split into small, non-contiguous blocks, none large enough for a new request

Explanation: External fragmentation happens with variable-size allocation (segmentation) where total free memory may be sufficient but is scattered in small non-contiguous chunks.

Q31 Which page replacement algorithm suffers from Belady's Anomaly?

ADVANCED

- A. FIFO**
- B. LRU
- C. Optimal
- D. Second Chance

Correct Answer: A. FIFO

Explanation: FIFO can exhibit Belady's Anomaly, where increasing the number of page frames actually increases the number of page faults for certain reference strings.

Q32 The Optimal page replacement algorithm replaces the page that:

INTERMEDIATE

- A. Will not be used for the longest time in the future**
- B. Was used least recently
- C. Was loaded first
- D. Has the smallest page number

Correct Answer: A. Will not be used for the longest time in the future

Explanation: Optimal (also called Belady's algorithm/MIN) replaces the page whose next use is farthest in the future — it's theoretical since future access isn't knowable in practice.

Q33 What is thrashing?

INTERMEDIATE

- A. A state where the system spends more time swapping pages than executing processes**
- B. A CPU running at 100% utilization efficiently
- C. A deadlock between two processes
- D. A disk running out of space

Correct Answer: A. A state where the system spends more time swapping pages than executing processes

Explanation: Thrashing occurs when the degree of multiprogramming is too high relative to available memory, causing excessive paging and very low actual CPU utilization on real work.

Q34 Which is TRUE about paging vs segmentation?

BASIC

- A. Paging divides memory into fixed-size blocks; segmentation divides it into variable-sized logical units**
- B. Paging causes external fragmentation; segmentation causes internal fragmentation
- C. Segmentation is always faster than paging
- D. Paging requires no page table

Correct Answer: A. Paging divides memory into fixed-size blocks; segmentation divides it into variable-sized logical units

Explanation: Paging uses fixed-size pages/frames (risking internal fragmentation); segmentation uses variable-size logical segments matching program structure (risking external fragmentation).

Q35 A Translation Lookaside Buffer (TLB) is used to:

INTERMEDIATE

- A. Cache recent virtual-to-physical address translations for faster memory access**
- B. Store the entire page table in the CPU
- C. Cache disk blocks
- D. Translate machine code to assembly

Correct Answer: A. Cache recent virtual-to-physical address translations for faster memory access

Explanation: The TLB is a small, fast cache of recent page-table entries, avoiding a full page table lookup (which could otherwise require an extra memory access) on every reference.

Q36 What does 'demand paging' mean?

BASIC

- A. Pages are loaded into memory only when they are actually referenced/needed**
- B. All pages are loaded at process start
- C. Pages are loaded in a fixed order
- D. Memory is never freed

Correct Answer: A. Pages are loaded into memory only when they are actually referenced/needed

Explanation: Demand paging is a lazy-loading strategy — a page is brought into memory only on a page fault when it's actually accessed, saving memory and startup time.

Q37 In a multi-level page table, the main advantage is:

ADVANCED

- A. It reduces memory overhead by not needing contiguous storage for a huge single-level page table**
- B. It eliminates the need for a TLB
- C. It removes internal fragmentation entirely
- D. It speeds up disk access

Correct Answer: A. It reduces memory overhead by not needing contiguous storage for a huge single-level page table

Explanation: A multi-level (hierarchical) page table avoids allocating a massive contiguous page table for sparse address spaces, since unused regions don't need table entries.

Q38 What triggers a page fault?

BASIC

- A. Accessing a page that is not currently present in physical memory**
- B. A process exceeding its time quantum
- C. A deadlock occurring
- D. A disk read error

Correct Answer: A. Accessing a page that is not currently present in physical memory

Explanation: A page fault is a trap raised by the MMU when a program accesses a page marked as not-present in the page table, requiring the OS to fetch it from disk.

5. File Systems, I/O & Disk Scheduling File allocation, directory structures, disk scheduling

algorithms

Q39 Which disk scheduling algorithm services requests strictly in the order they arrive? **BASIC**

- A. FCFS**
- B. SSTF
- C. SCAN
- D. C-LOOK

Correct Answer: A. FCFS

Explanation: FCFS (First Come First Served) disk scheduling processes I/O requests in the exact order received, without reordering for efficiency.

Q40 SSTF disk scheduling stands for and prioritizes:

INTERMEDIATE

- A. Shortest Seek Time First — services the request closest to the current head position**
- B. Slowest Seek Time First
- C. Sequential Sector Track Fetch
- D. Static Sector Time Formatting

Correct Answer: A. Shortest Seek Time First — services the request closest to the current head position

Explanation: SSTF selects the pending request with the minimum seek distance from the current head position, reducing total seek time but risking starvation of far requests.

Q41 The SCAN disk scheduling algorithm is also known as:

INTERMEDIATE

- A. The elevator algorithm**
- B. The shortest job algorithm
- C. The banker's algorithm
- D. The round robin algorithm

Correct Answer: A. The elevator algorithm

Explanation: SCAN moves the disk arm in one direction servicing requests, reverses at the end — just like an elevator moving up/down servicing floors.

Q42 Which file allocation method suffers most from external fragmentation?

INTERMEDIATE

- A. Contiguous allocation**
- B. Linked allocation
- C. Indexed allocation
- D. FAT-based allocation

Correct Answer: A. Contiguous allocation

Explanation: Contiguous allocation requires each file to occupy a contiguous block of disk space, leading to external fragmentation as files are created/deleted over time.

Q43 In linked file allocation, a major drawback is:

INTERMEDIATE

- A. Slow random/direct access since blocks must be traversed sequentially via pointers**
- B. High internal fragmentation
- C. It cannot support large files
- D. It requires contiguous free space

Correct Answer: A. Slow random/direct access since blocks must be traversed sequentially via pointers

Explanation: Since each block points to the next, accessing a specific block (e.g., block 50) requires traversing all preceding blocks sequentially — direct access is inefficient.

Q44 An inode in a UNIX-like file system stores:

ADVANCED

- A. Metadata about a file (permissions, size, pointers to data blocks) but not the filename**
- B. Only the filename
- C. The entire file content
- D. Only the file's creation date

Correct Answer: A. Metadata about a file (permissions, size, pointers to data blocks) but not the filename

Explanation: An inode holds file metadata and data-block pointers; the filename-to-inode mapping is stored separately in the directory entry.

Q45 What is the purpose of a superblock in a file system?

ADVANCED

- A. It stores metadata about the file system itself (size, block count, free blocks)**
- B. It stores the operating system kernel
- C. It is the root directory's inode
- D. It caches recently accessed files

Correct Answer: A. It stores metadata about the file system itself (size, block count, free blocks)

Explanation: The superblock contains file-system-wide metadata such as total blocks, free block count, and file system type — critical for mounting the file system.

Q46 Which of the following best describes 'buffering' in I/O management?

BASIC

- A. Temporarily storing data in memory to smooth out speed differences between devices**
- B. Permanently storing files on disk
- C. Encrypting I/O data
- D. Compressing files before storage

Correct Answer: A. Temporarily storing data in memory to smooth out speed differences between devices

Explanation: Buffering uses memory areas to hold data being transferred, compensating for speed mismatches between fast CPU/memory and slower I/O devices.

6. Threads, Concurrency & Advanced Topics Multithreading models, real-time OS,

misc advanced

Q47 What is the main advantage of multithreading over multiprocessing?

BASIC

- A. Threads share the same memory space, making context switching and communication cheaper**
- B. Threads always run faster on single-core CPUs
- C. Threads cannot crash the whole process
- D. Threads require no synchronization

Correct Answer: A. Threads share the same memory space, making context switching and communication cheaper

Explanation: Since threads within a process share an address space, creating threads and switching between them is cheaper than processes, and inter-thread communication is direct via shared memory.

Q48 In the Many-to-One threading model:

ADVANCED

- A. Many user-level threads map to a single kernel thread**
- B. One user thread maps to one kernel thread
- C. Many user threads map to many kernel threads flexibly
- D. There is no kernel thread involved

Correct Answer: A. Many user-level threads map to a single kernel thread

Explanation: Many-to-One maps multiple user threads to a single kernel thread; efficient for management but a blocking system call blocks the entire process since the kernel sees only one thread.

Q49 The One-to-One threading model (e.g., in modern Linux/Windows) offers:

ADVANCED

- A. True parallelism since each user thread maps to its own kernel thread**
- B. No parallelism at all
- C. Fewer kernel resources used than many-to-one
- D. Only cooperative scheduling

Correct Answer: A. True parallelism since each user thread maps to its own kernel thread

Explanation: One-to-One models allow multiple threads to run in parallel on multicore systems since each has a dedicated kernel thread, though it has more overhead creating each thread.

Q50 What distinguishes a Real-Time Operating System (RTOS) from a general-purpose OS?

INTERMEDIATE

- A. It guarantees processing within strict, predictable time constraints (deadlines)**
- B. It is always faster in raw throughput
- C. It has no scheduler
- D. It cannot run multiple processes

Correct Answer: A. It guarantees processing within strict, predictable time constraints (deadlines)

Explanation: RTOS prioritizes deterministic, predictable response times to meet hard or soft deadlines, sometimes at the cost of overall throughput compared to general-purpose OS.

Q51 A hard real-time system is one where:

ADVANCED

- A. Missing a deadline is considered a total system failure**
- B. Missing a deadline slightly degrades performance but is tolerable
- C. There are no deadlines
- D. Deadlines are optional

Correct Answer: A. Missing a deadline is considered a total system failure

Explanation: In hard real-time systems (e.g., pacemakers, aircraft controls), missing even one deadline can be catastrophic; soft real-time systems tolerate occasional missed deadlines.

Q52 What is a monitor in the context of process synchronization?

ADVANCED

- A. A high-level synchronization construct that encapsulates shared data with mutual-exclusion access via procedures**
- B. A hardware display device
- C. A type of process scheduler
- D. A page replacement algorithm

Correct Answer: A. A high-level synchronization construct that encapsulates shared data with mutual-exclusion access via procedures

Explanation: A monitor is a programming-language-level construct that bundles shared variables and the procedures that operate on them, automatically ensuring mutual exclusion.

Q53 Which of these best describes 'system calls'?

BASIC

- A. An interface allowing user programs to request services from the OS kernel**
- B. A hardware interrupt only
- C. A type of process scheduling
- D. A file compression technique

Correct Answer: A. An interface allowing user programs to request services from the OS kernel

Explanation: System calls (e.g., read, write, fork, exec) provide the programmatic interface for user-space applications to request kernel-level services.